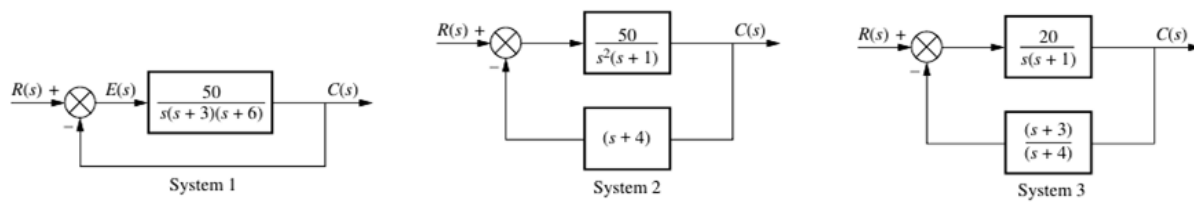


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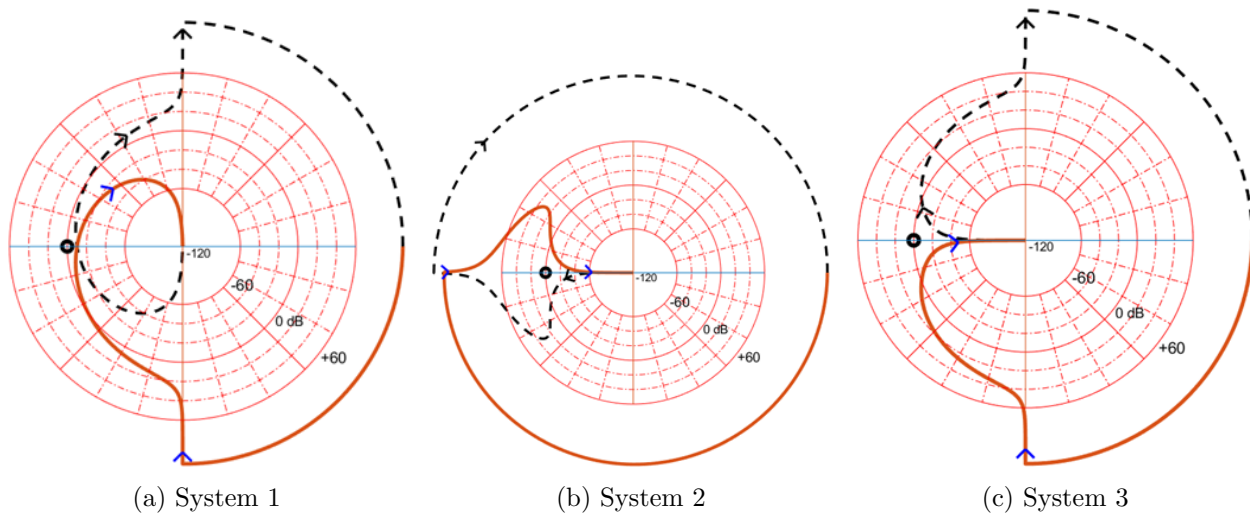
EEE3094S - Control Systems Engineering

Problem Set 5.2: Closed Loop Stability Analysis

Question 1



The systems above have the following Nyquist plots: (Black circle = -1)

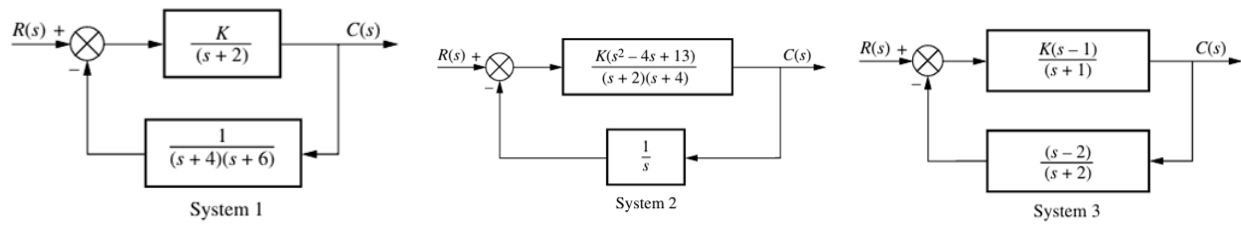


Using the Nyquist stability criterion, determine whether each system is stable.

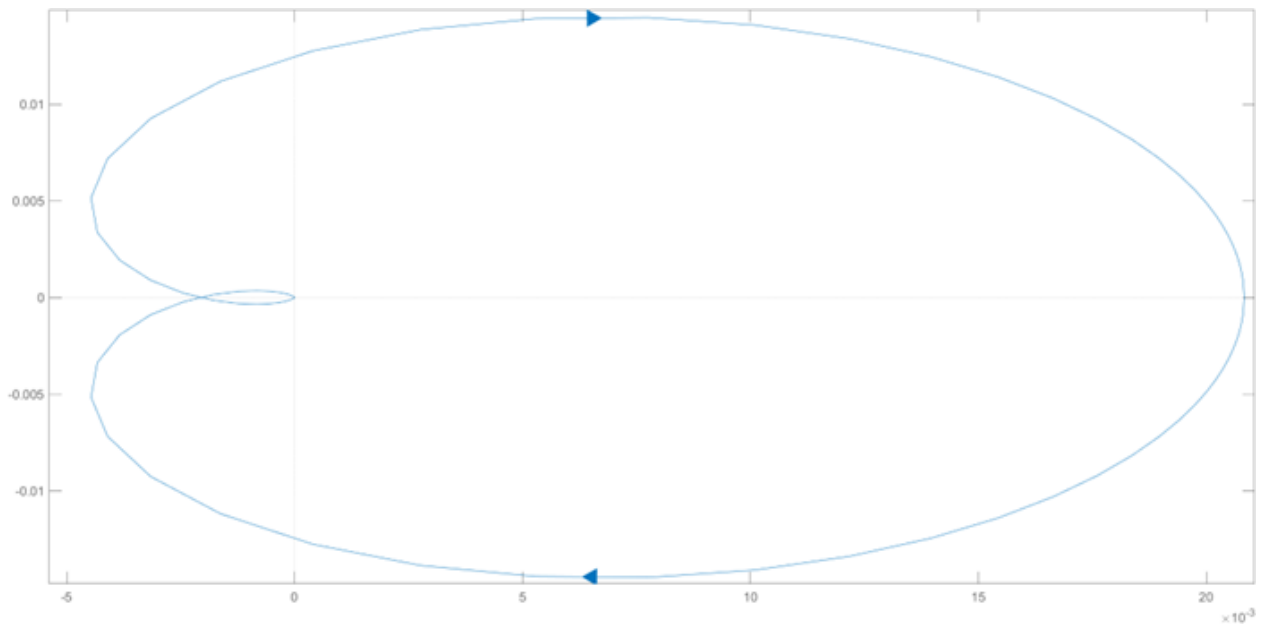
Solution:

- a. $P = 0$, $N = 0$. Therefore, $Z = P - N = 0$. System is **stable**.
- b. $P = 0$, $N = -2$. Therefore, $Z = P - N = 2$. System is **unstable**.
- c. $P = 0$, $N = 0$. Therefore, $Z = P - N = 0$. System is **stable**.

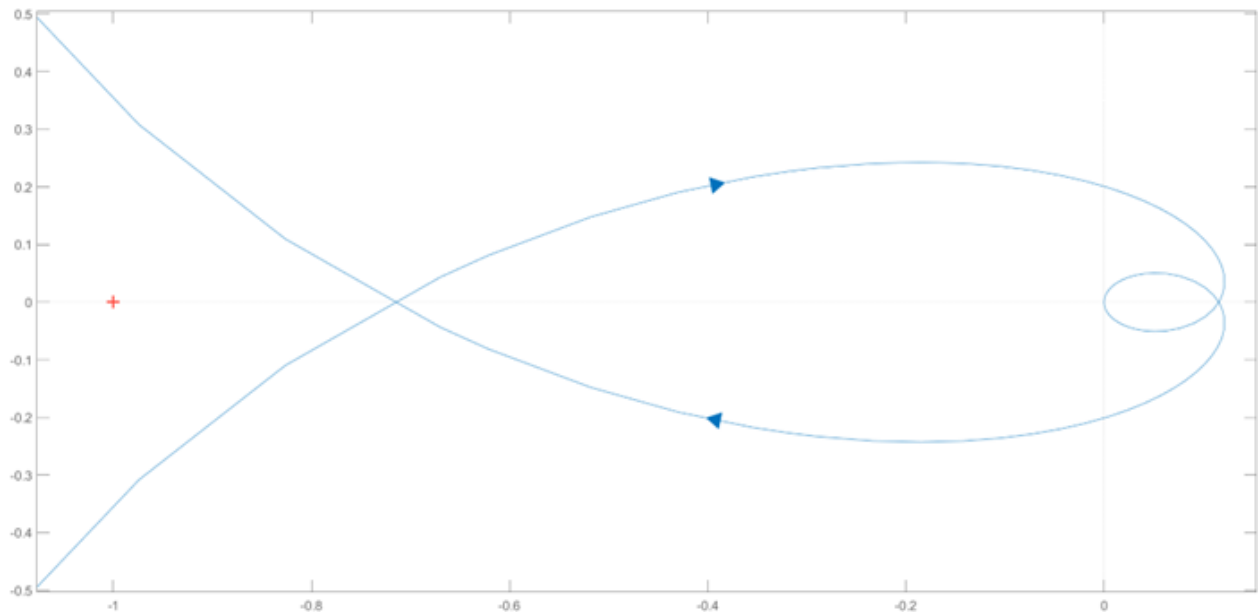
Question 2



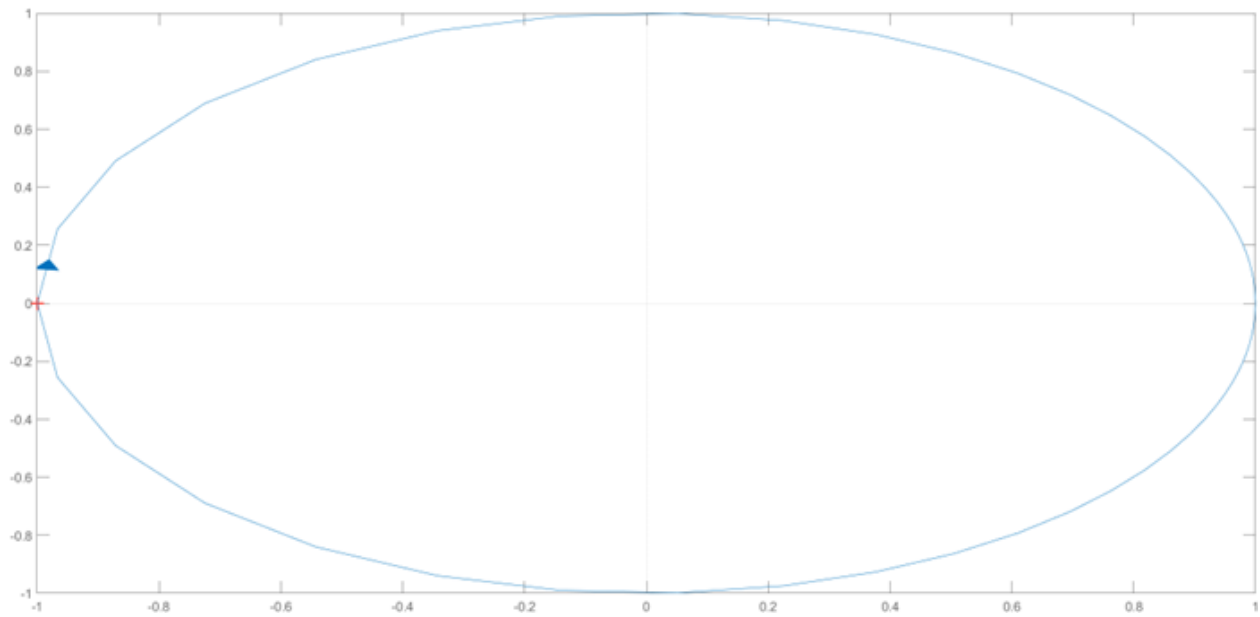
The systems above have the following Nyquist plots for unity gain ($K = 1$):



System 1



System 2



System 3

Using the Nyquist stability criterion, find the range of K values for which each system is stable.

Solution:

1. The Nyquist diagram intersects the real axis at -0.0021 . Thus, K can be increased to 478.63 before there are encirclements of -1 . Since no poles are encircled, $P = 0$. Therefore,

$$Z = P - N = \begin{cases} 0 - 0 = 0 & \text{if } 0 < K < 478.63 \Rightarrow \text{Stable,} \\ 0 - (-2) = 2 & \text{if } K > 478.63 \Rightarrow \text{Unstable.} \end{cases}$$

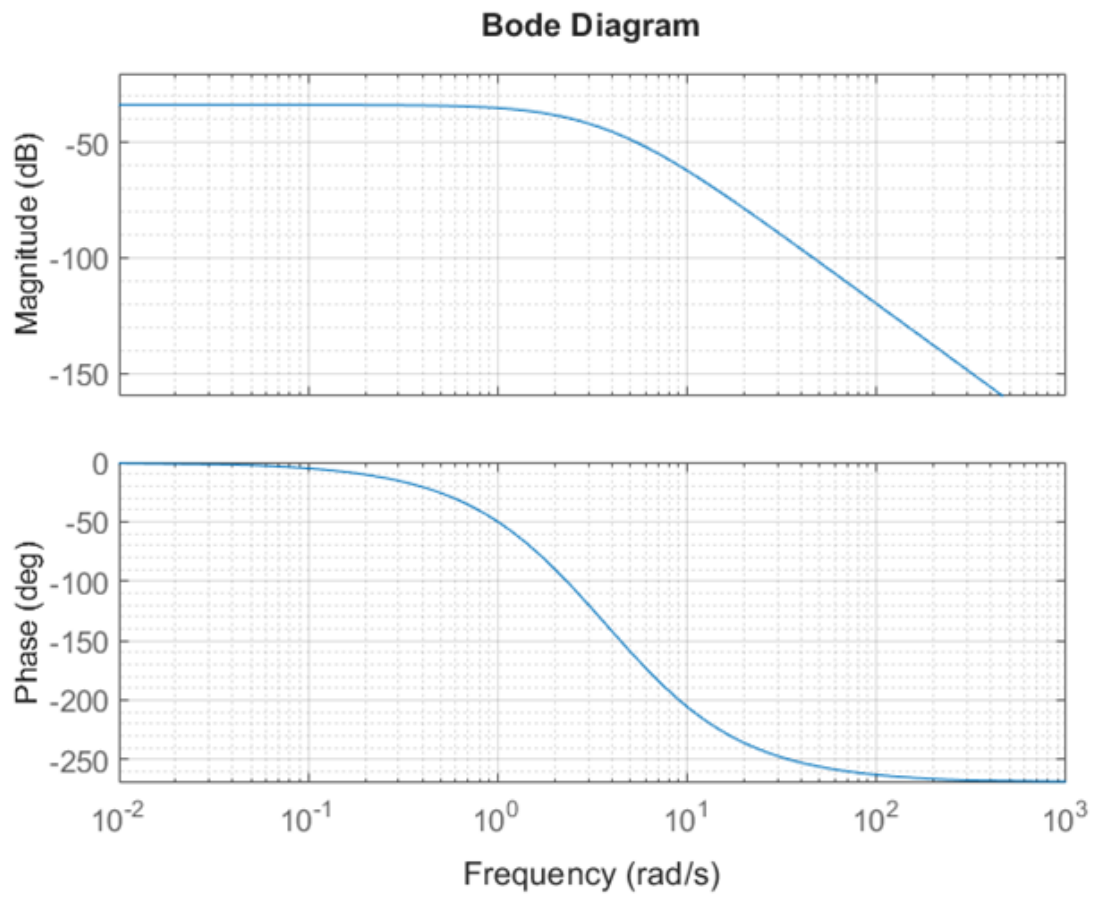
2. The Nyquist diagram intersects the real axis at -0.720 . Thus, K can be increased to 1.39 before there are encirclements of -1 . With $P = 0$, we have

$$Z = P - N = \begin{cases} 0 - 0 = 0 & \text{if } 0 < K < 1.39 \Rightarrow \text{Stable,} \\ 0 - (-2) = 2 & \text{if } K > 1.39 \Rightarrow \text{Unstable.} \end{cases}$$

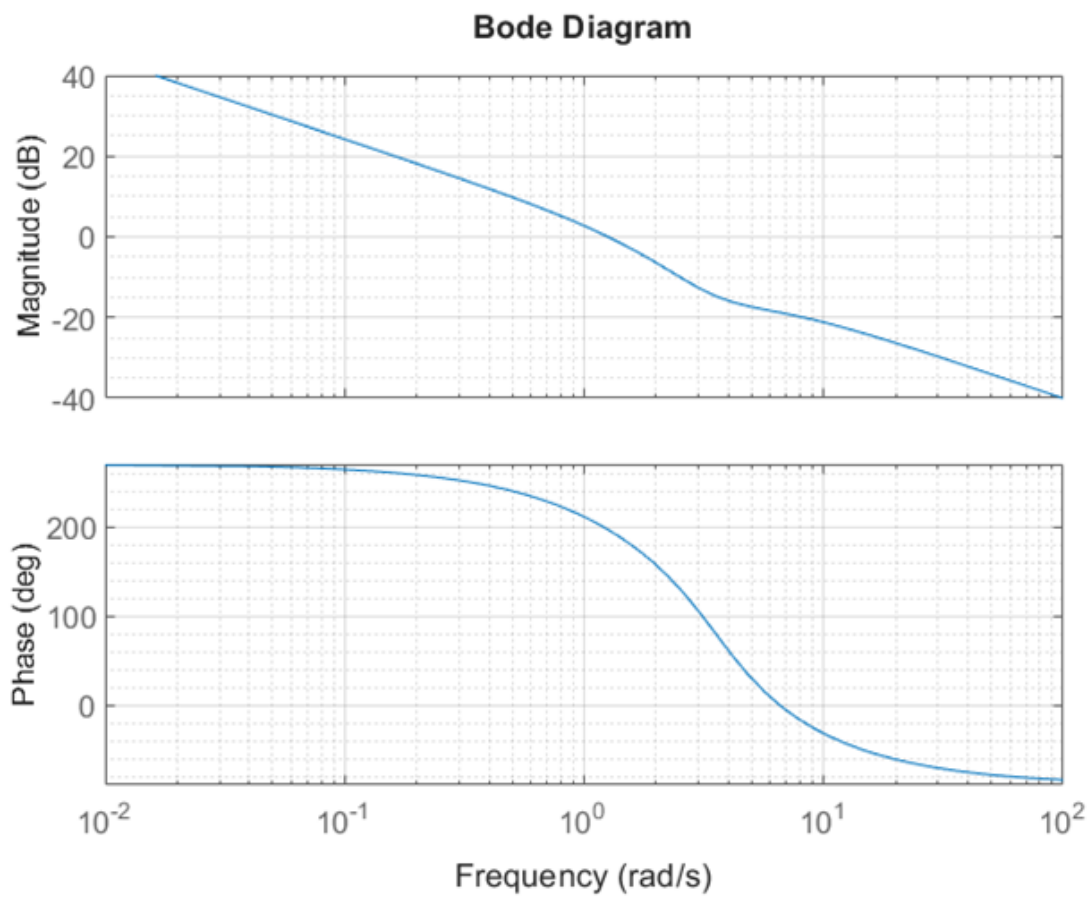
3. The system is **stable** if $0 < K < 1$.

Question 3

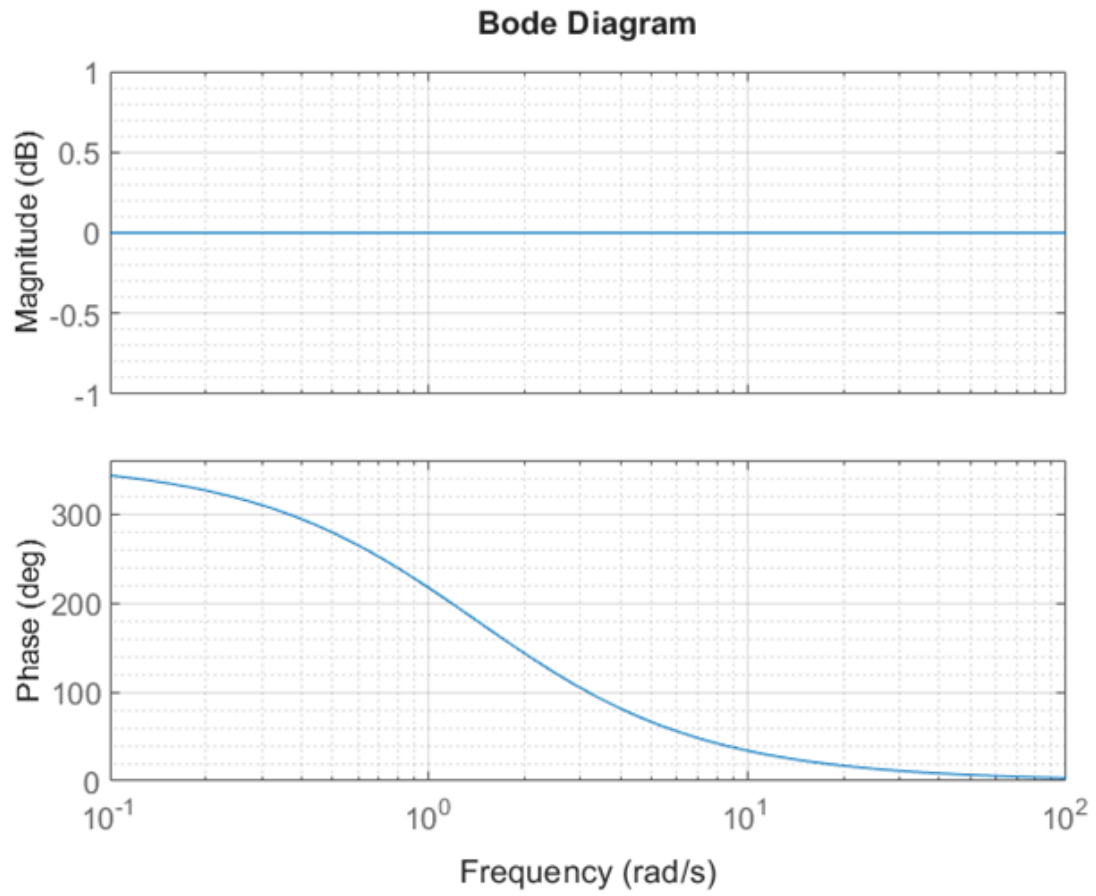
The systems in Question 2 have the following open-loop frequency responses for unity gain:



System 1



System 2



System 3

For each system in Question 2, find the gain margin and phase margin if the value of K is:

- i) $K = 1000$
- ii) $K = 100$
- iii) $K = 0.1$

Solution:

System 1

- $K = 1000$:

For $K = 1$, phase is 180° at $\omega = 6.63 \text{ rad/s}$ and magnitude is -53.6 dB .

For $K = 1000$, the magnitude curve shifts up by 60 dB to $+6.4 \text{ dB}$ at $6.63 \text{ rad/s} \Rightarrow$ **Gain margin** $= -6.4 \text{ dB}$.

Phase margin: the 0 dB crossing occurs at $\omega = 9.07 \text{ rad/s}$ where phase is 200.3° . Hence **Phase margin** $= 180^\circ - 200.3^\circ = -20.3^\circ$.

- $K = 100$:

For $K = 1$, phase is 180° at $\omega = 6.63 \text{ rad/s}$ and magnitude is -53.6 dB .

For $K = 100$, the magnitude curve shifts up by 40 dB to -13.6 dB at $6.63 \text{ rad/s} \Rightarrow$ **Gain margin** $= 13.6 \text{ dB}$.

Phase margin: the 0 dB crossing occurs at $\omega = 2.54 \text{ rad/s}$ where phase is 107.3° . Hence **Phase margin** $= 180^\circ - 107.3^\circ = 72.7^\circ$.

- $K = 0.1$:

For $K = 1$, phase is 180° at $\omega = 6.63 \text{ rad/s}$ and magnitude is -53.6 dB .

For $K = 0.1$, the magnitude curve shifts down by 20 dB to -73.6 dB at $6.63 \text{ rad/s} \Rightarrow$ **Gain margin** $= 73.6 \text{ dB}$.

System 2

- $K = 1000$:

For $K = 1$, phase is 180° at $\omega = 1.56 \text{ rad/s}$ and magnitude is -2.85 dB .

For $K = 1000$, the magnitude curve shifts up by 60 dB to $+57.15 \text{ dB}$ at $1.56 \text{ rad/s} \Rightarrow$ **Gain margin** $= -57.15 \text{ dB}$.

Phase margin: raising by 54 dB gives 0 dB at $\omega = 500 \text{ rad/s}$ where phase is -91.03° . Hence **Phase margin** $= 180^\circ - 91.03^\circ = 88.97^\circ$.

- $K = 100$:

For $K = 1$, phase is 180° at $\omega = 1.56 \text{ rad/s}$ and magnitude is -2.85 dB .

For $K = 100$, the magnitude curve shifts up by 40 dB to $+37.15 \text{ dB}$ at $1.56 \text{ rad/s} \Rightarrow$ **Gain margin** $= -37.15 \text{ dB}$.

Phase margin: the 0 dB crossing occurs at $\omega = 99.8 \text{ rad/s}$ where phase is -84.3° . Hence **Phase margin** $= 180^\circ - 84.3^\circ = 95.7^\circ$.

- $K = 0.1$:

For $K = 1$, phase is 180° at $\omega = 1.56 \text{ rad/s}$ and magnitude is -2.85 dB .

For $K = 0.1$, the magnitude curve shifts down by 20 dB to -22.85 dB at $1.56 \text{ rad/s} \Rightarrow$ **Gain margin** $= -22.85 \text{ dB}$.

Phase margin: the 0 dB crossing occurs at $\omega = 0.162 \text{ rad/s}$ where phase is -99.86° . Hence **Phase margin** $= 180^\circ - 99.86^\circ = 80.2^\circ$.

System 3

- $K = 1000$:

For $K = 1$, phase is 180° at $\omega = 1.41$ rad/s and magnitude is 0 dB.

For $K = 1000$, the magnitude curve shifts up by 60 dB to 60 dB at 1.41 rad/s $\Rightarrow$ **Gain margin** = -60 dB.

Phase margin: no 0 dB crossing occurs $\Rightarrow$ **Phase margin** = ∞ .

- $K = 100$:

For $K = 1$, phase is 180° at $\omega = 1.41$ rad/s and magnitude is 0 dB.

For $K = 100$, the magnitude curve shifts up by 40 dB to 40 dB at 1.41 rad/s $\Rightarrow$ **Gain margin** = -40 dB.

Phase margin: no 0 dB crossing occurs $\Rightarrow$ **Phase margin** = ∞ .

- $K = 0.1$:

For $K = 1$, phase is 180° at $\omega = 1.41$ rad/s and magnitude is 0 dB.

For $K = 0.1$, the magnitude curve shifts down by 20 dB to -20 dB at 1.41 rad/s $\Rightarrow$ **Gain margin** = 20 dB.

Phase margin: no 0 dB crossing occurs $\Rightarrow$ **Phase margin** = ∞ .